

Integrated Website Capstone Project

Project title

“Build a Complete Small-Business Website”

Students work individually or in pairs to design and develop a functional website for a fictional or local business.

Examples:

- Bookstore 📖
- Café ☕
- Clothing boutique 👗
- Travel agency ✈️
- Digital academy 🎓
- Restaurant 🍽️
- Beauty salon 💇
- Electronics shop 📱
- Sports store 🏃
- Small hotel 🏨
- Freelance agency 💻

The project should assess:

Skill	Assessment focus
HTML	Semantic structure, forms, tables, navigation, accessibility
CSS	Layout, responsive design, typography, colors, spacing, UI
JavaScript	DOM manipulation, validation, interaction, calculations/filtering
SQL	Database design, tables, relationships, CRUD/query operations
Canva	Brand identity, wireframe/mockup, promotional/social media material

1. Project Scenario

Give students this scenario:

You have been hired as a junior web development team to create an online presence for a small business. The client needs a professional website that presents the business, its products/services, and allows visitors to interact with the website. The business also needs a database to manage relevant information.

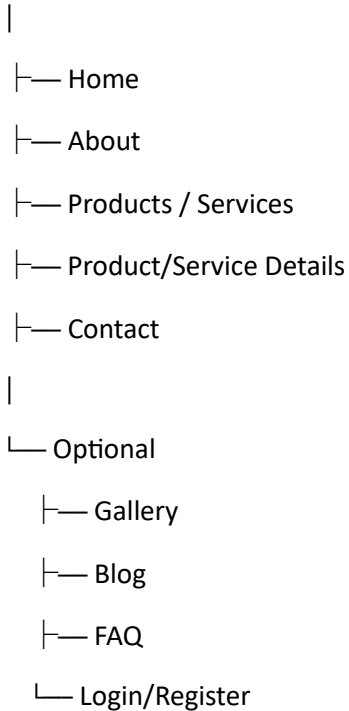
This gives them a realistic client-style project rather than simply saying “*make a website.*”

2. Required Website

Every student/team must produce a website with **at least 5 pages**.

Recommended structure

Business Website



I would make **Home, About, Products/Services and Contact mandatory**, while allowing the fifth page to depend on their business.

3. HTML Assessment

Students should demonstrate the following.

A. Document structure

They must correctly use:

`<!DOCTYPE html>`

`<html>`

`<head>`

`<body>`

and appropriate metadata.

B. Semantic HTML

Require appropriate use of:

<header>

<nav>

<main>

<section>

<article>

<aside>

<footer>

rather than building everything with <div>.

C. Navigation

The website must have:

- Logo/business name
- Navigation menu
- Internal page links
- Active/current page indication

D. Content

Students should demonstrate:

- Headings
- Paragraphs
- Lists
- Images
- Links
- Tables where appropriate
- Buttons
- Cards

E. Forms

The Contact page should contain a proper form:

Name

Email

Phone

Subject

Message

Submit

Reset

Additional fields can be business-specific.

F. HTML quality

Assess:

- Proper nesting
 - Meaningful alt attributes
 - Labels associated with form controls
 - Appropriate input types
 - Clean indentation
 - No unnecessary tags
-

4. CSS Assessment

This is where you can differentiate between a merely functional website and a professionally designed one.

Required CSS concepts

Layout

Students must demonstrate at least:

- Flexbox
- CSS Grid
- Box model
- Width/height
- Margin
- Padding
- Positioning

Typography

They should control:

- Font family
- Font size
- Font weight

- Line height
- Text alignment

Visual design

Require:

- Consistent color palette
- Buttons
- Cards
- Navigation styling
- Borders
- Border radius
- Shadows where appropriate
- Hover effects

Responsive design

This should be **mandatory**.

The website must work reasonably on:

Desktop

Tablet

Mobile

Students should use media queries.

For example:

```
@media (max-width: 768px) {  
  /* responsive styles */  
}
```

CSS organization

Assess whether students understand the difference between:

HTML = structure

CSS = presentation

JavaScript = behavior

I would specifically discourage putting large amounts of inline CSS in the project.

5. JavaScript Assessment

JavaScript should not be reduced to a simple alert box.

Give them **minimum functional requirements**.

Required JavaScript features

1. Form validation

Contact form should validate:

- Required fields
- Email format
- Minimum message length
- Phone number format where applicable

2. DOM manipulation

At least one feature should dynamically change the page.

For example:

Add to Cart

Remove Item

Change Quantity

Show/Hide Information

Change Theme

Display Message

3. Interactive navigation

For example:

- Mobile menu
- Dropdown menu
- FAQ accordion
- Modal
- Tabs

4. Search/filter

On the Products page:

Search products



JavaScript



Display matching products

or category filtering.

5. Calculation

Require at least one meaningful calculation.

Examples:

Shopping website

Price × Quantity = Subtotal

Restaurant

Item price × quantity

+ tax

= Total

Travel agency

Package price × number of people

This makes JavaScript much more meaningful.

6. SQL / Database Component

This is where I would turn the project into a genuine **full-stack foundation**.

Students should design a database appropriate to their website.

For example, a bookstore might have:

bookstore_db

Customers

Products

Categories

Orders

Order_Details

Messages

Minimum requirement

Every project should have:

- Database
- At least **3 related tables**

- Primary keys
- Foreign keys
- Appropriate data types

Example

categories

category_id PK

category_name

products

product_id PK

category_id FK

product_name

price

description

stock

image

customers

customer_id PK

name

email

phone

7. SQL Query Requirements

Students should submit SQL demonstrating:

Database operations

CREATE DATABASE

CREATE TABLE

ALTER TABLE

DROP TABLE

Data operations

INSERT

SELECT

UPDATE

DELETE

Conditions

WHERE

AND

OR

LIKE

BETWEEN

IN

Sorting

ORDER BY

Aggregation

At least:

COUNT()

SUM()

AVG()

MAX()

MIN()

Example business questions

Instead of simply asking them to write queries, give them questions such as:

Find all products belonging to the "Books" category.

Find customers who have placed orders.

Calculate total sales.

Find the most expensive product.

Count products in each category.

That assesses whether they can actually **use SQL to answer business questions**.

8. Canva Component 🎨

I would deliberately include Canva because it assesses something students often overlook: **visual communication and client presentation**.

Students must create a small **brand kit** for their website.

Canva Deliverable 1 — Logo

Create:

- Business logo
- Icon/mark if appropriate
- Horizontal version if appropriate

Canva Deliverable 2 — Brand Board

One page containing:

Business Name

Logo

Color Palette

Primary Font

Secondary Font

Brand Style

Short Brand Statement

Canva Deliverable 3 — Website Wireframe

Students create a visual wireframe/mockup of their:

- Home page
- One internal page

This is particularly useful because you can compare:

Canva design → actual HTML/CSS implementation

and assess whether they can translate a visual design into code.

Canva Deliverable 4 — Social Media Post

Create one promotional graphic:

Instagram/Facebook promotional post

Example:

20% OFF

Summer Collection

Shop Now

This gives you a small but useful introduction to **digital marketing/design thinking**.

9. Project Folder Structure

Make folder organization part of the assessment.

I recommend requiring:

business-website/

|

|— index.html

|— about.html

|— products.html

|— details.html

|— contact.html

|

|— css/

| └─ style.css

|

|— js/

| └─ script.js

|

|— images/

| └─ logo.png

| └─ products/

| └─ banners/

|

|— database/

| └─ database.sql

|

- └─ canva/
 - | └─ brand-board.pdf
 - | └─ wireframe.pdf
 - | └─ social-post.png

This also quietly assesses their **professional project management habits**.
